

Hackathon Problem Statements

Regd No: 25091A32G0, 25091A32M0

Problem Statement No: 15

Multilingual Crop Advisory Chatbot Backend Develop the backend logic for a crop advisory chatbot that uses basic NLP techniques to parse farmer queries in regional languages and route them using a priority queue based on urgency (e.g., pest attack vs general query). Integrate a knowledge base built from agricultural extension datasets to generate rule-based responses. Use OOP to structure different advisory modules (irrigation, pest, fertilizer) as interchangeable strategy classes. This bridges the information gap for farmers without direct access to agricultural experts.

Regd No: 25091A3220, 25091A3265

Problem Statement No: 52

Encroachment Detection Using GIS Data Classification Build a basic land encroachment detection tool using classification techniques on simulated or public satellite/GIS boundary datasets to flag mismatches between registered land boundaries and actual land use. Use OOP to model land parcels, ownership records, and encroachment flags. This supports municipal survey departments in identifying unauthorized construction and land disputes more systematically.

Regd No: 25091A32G7, 25091A32E8

Problem Statement No: 91

Microfinance Loan Default Risk Predictor Build a loan default risk prediction model using classification algorithms trained on microfinance borrower datasets (repayment history, income proxy indicators, loan purpose) to help microfinance institutions make more informed lending decisions. Use OOP to model borrowers, loans, and risk profiles. This supports financial inclusion by helping lenders extend credit responsibly to underserved populations without excessive collateral requirements.

Regd No: 25091A3207, 25091A3213

Problem Statement No: 122

Government Scheme Eligibility Checker Design an eligibility checking tool using an OOP-based rule engine that evaluates a citizen's profile against multiple government welfare scheme criteria simultaneously (income, category, age, occupation) to recommend all schemes they qualify for. This addresses the widespread problem of eligible citizens missing out on benefits simply due to lack of awareness about complex, fragmented scheme criteria.

Regd No: 24091A3261, 25091A32G6

Problem Statement No: 131

Migrant Worker Registration & Welfare Benefit Tracker Build a migrant worker welfare tracking system using an OOP inheritance hierarchy to model different worker categories and their eligible benefits (healthcare, housing, education for children) across state boundaries. This addresses the well-documented gap in welfare access faced by inter-state migrant workers who fall through jurisdictional cracks in benefit delivery systems.

Regd No: 25091A32E4, 25091A32A6

Problem Statement No: 144

Fake News/Misinformation Classifier Create a misinformation detection tool using NLP-based text feature analysis (source credibility signals, linguistic patterns, sensationalism markers) trained on public fake-news datasets to classify news articles as likely reliable or unreliable. Use OOP to model articles, sources, and classification confidence scores. This addresses the growing societal challenge of misinformation spread, particularly relevant during elections and public health crises.